

GE Healthcare

GE 6-Channel Phased Array Flex Coil Operator Manual



GE 6-Channel Phased Array Flex Coil
Operator Manual, English
5371823-2EN
Revision 1
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Medical Device Directive

This product conforms with the requirements of council directive 93/42/EEC concerning medical devices and bears the following CE Conformity Marking:



European Representative:

GE Medical Systems S.C.S
Quality Assurance Manager
283 rue de la Minière
78530 BUC France
Telephone: +33 1 30 70 40 40

Do not use devices that transmit RF Signals (**cellular phones**, transceivers, or radio controlled products) in the vicinity of this equipment as they may cause performance outside the published specifications. Keep the power to these types of devices turned off when near this equipment.

The medical staff in charge of this equipment is required to instruct technicians, patients, and other people who may be around this equipment to fully comply with the above requirement.

Immunity/Emissions Exceptions: Note the exceptions from the EMC test results. Check with the business EMC engineer for this information.

In accordance with the international safety standard IEC 60601-1, this system is a Class II device, acceptable for Continuous Operation, having ordinary protection against ingress of water (IPX0) with type BF applied parts and is not for use in the presence of flammable anesthetics.





CAUTION: This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.

Table of Contents

Chapter 1: Safety

Introduction	1-1
Use of symbols	1-2
Indications for use	1-3
Coil operating safety	1-4
Quality assurance.....	1-4
Coil modifications	1-4
Patient safety	1-5
Emergency procedures.....	1-5
Patient contraindications.....	1-5
Equipment safety	1-6
Electrical and mechanical safety	1-6
Incident reporting	1-7

Chapter 2: GE 6-Channel Phased Array Flex Coil

Introduction	2-1
Shipping contents	2-2
Coil description.....	2-3
Intended uses	2-3
Coil components	2-3
Coil positioning.....	2-4
Routine coil positioning procedures	2-4
Patient positioning.....	2-5
Modes of operation	2-9

Chapter 3: Maintenance

Introduction	3-1
Coil and components care	3-2
Cleaning.....	3-2
Storage.....	3-2

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Chapter 1

Safety

Introduction

This chapter describes the safety precautions of the **GE 6-Channel Phased Array Flex Coil**, compatible with the GE 1.5T MR450, MR450w, and 3.0T MR750 whole body MRI systems. This manual contains detailed information on the set-up, positioning, and use of your coil. The instructions should be read carefully and thoroughly before attempting to scan patients with the coil.

The coil is indicated for use, on the order of a physician in conjunction with an MR scanner, as an accessory to produce 2D and 3D images.

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to prevent patient exposure to electrical or mechanical hazards that may cause injury.

If you have any questions or comments on this manual, or need any assistance with the use of the product, please contact your GE Healthcare representative at: 1-800-GE CARES.



CAUTION: Federal law restricts this device to sale, distribution, and use by or on the order of a physician.

You should be familiar with the following safety guidelines before using the **GE 6-Channel Phased Array Flex Coil**:

- [Use of symbols](#)
- [Indications for use](#)
- [Coil operating safety](#)
- [Patient safety](#)
- [Equipment safety](#)
- [Electrical and mechanical safety](#)
- [Incident reporting](#)

Use of symbols

The following safety notices are used to emphasize certain safety instructions. This manual uses the international symbol along with the danger, warning, or caution message.



DANGER: Danger is used to conditions or actions for which a specific hazard is known to exist which will cause severe personal injury, death, or substantial property damage if the instructions are ignored.



WARNING: Warning is used to identify conditions or actions for which a specific hazard is known to exist which may cause severe personal injury, death, or substantial property damage if the instructions are ignored.



CAUTION: Caution is used to identify conditions or actions for which a potential hazard may exist which will or can cause minor personal injury or property damage if the instructions are ignored.



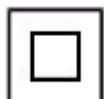
Coil CAUTION: Coil Caution is used to identify conditions or actions for which a potential hazard of crossing or looping coil cables may exist which will or can cause minor personal injury or property damage if the instructions are ignored.

IMPORTANT: Important indicates information where adherence to procedures is crucial or where your comprehension is necessary to apply a concept or effectively use the product.

NOTE: A Note provides additional information that is helpful to you. It may emphasize certain information regarding special tools or techniques, items to check before proceeding, or factors to consider about a concept or task.

One or more of the product identification labels may be on your imaging coils. Please familiarize yourself with the labels that apply to your particular peripheral equipment.

Table 1-1 Product identification labels

Label	Description (typical use)
	Class II Equipment (double insulated) (ratings)
	Type BF Applied Part
	Attention – Consult accompanying documents

Indications for use

The **GE 1.5T 6-Channel Phased Array Flex Coil** is indicated for use on the order of the physician in conjunction with a 1.5T MRI scanner, as an accessory to produce 2D and 3D images.

The **GE 3.0T 6-Channel Phased Array Flex Coil** is indicated for use on the order of the physician in conjunction with a 3.0T MRI scanner, as an accessory to produce 2D and 3D images.



CAUTION: This device is limited by federal law to investigational use for indications not in the “Indications for use” statement.



CAUTION: Federal law restricts the sale, distribution, and use of this device to or on the order of a physician.

Coil operating safety



WARNING: The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system operators manual.

- Cables should not be looped or crossed as arcing and patient burns could result.
- Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.
- Do not allow the surface coil cable to touch the patient; patient burns can result. Use a thermal resistant material or pad to keep the cable from touching the patient.
- Keep the length of the cable in the bore to a minimum. Avoid bending the cable 180 degrees and route the cables out of the bore in the most direct way - without looping or coiling.
- Visually inspect the cable insulator jackets, strain reliefs and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.
- Remove unplugged surface coils or unused accessory devices from the magnet bore; a patient burn can result.
- Remove blue wire-type ECG gating cables from the magnet bore before scanning with surface coils.
- Do not use blue wire-type ECG leads when you are doing gated surface coil scans. When they are used in the presence of a surface coil cable, the potential for heating increases. Use peripheral gating or high impedance ECG leads rather than blue wire-type ECG leads when you are doing gated surface coil scans.
- When using any coil, make sure the patient's body does not touch the magnet bore. Use thermal resistant material or padding between the patient and the magnet to prevent burns that could be caused by patient contact with the interior of the magnet bore.
- Use only GE or GE-authorized accessory coils and cables.
- Use only accessories in good condition. If you suspect that an accessory is not in good condition, discontinue its use and contact your local GE service engineer.
- Do not use surface coils with exposed metal conductors or damaged insulation. Skin contact with metal conductors can cause burns.

Quality assurance

The MCQA Setup and Test procedure, described in the GE 6-Channel Phased Array Flex Coil service manual (5371824), should be performed by a GE Field Service Engineer upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

Coil modifications

Any changes or modifications to the coil must be approved and performed by GE Healthcare prior to installation.

Patient safety



WARNING: The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system operators manual.

- If a patient complains of burning, tingling, stinging, or other similar "burn"-type sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.
- Ensure that the patient is comfortably positioned.
- Monitor the patient periodically. Stop the scan immediately if the patient reports any heating, burning or tingling sensations.



Coil CAUTION: Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Emergency procedures

In the unlikely event that a coil creates smoke, sparks, or makes an unusually loud noise, or if the patient requires emergency assistance:

- stop the scan if one is in progress
- remove the patient from the scan room

Patient contraindications



WARNING: Contraindications

- Patients with ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.
- Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.
- There is a risk to scanning feverish or decompensated cardiac patients.
- Facial makeup should be removed before scanning because it may contain metal flakes, which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.
- Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing an MR exam.

Equipment safety

All personnel using the system should be instructed in the proper operation and use of the system. Personnel should observe all warnings and cautions that appear in the manual.



WARNING: Auxiliary equipment (such as gating equipment, vital signs monitoring systems and RF coils) that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.



WARNING: The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen or nitrous oxide.



Coil CAUTION: Prevent cables from forming loops. Looping degrades scan performance of the coil through RF coupling.



WARNING: This equipment is not protected against ingress of water (IPX0).

NOTE: This equipment is rated for continuous operation.

NOTE: For the disposal of this coil at the end of its useful life, please contact the GE Service Personnel.

Electrical and mechanical safety



WARNING: The Coil consists of both electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it for any external damage. Do not use the coil if the housing or cable is broken.

Incident reporting

Contact GE Healthcare immediately to report an incident and/or injury to a patient, operator, or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

For additional information, contact:

GE Healthcare – Americas

- USA: 800-558-5102
- Canada: 800-668-0732

GE Healthcare – Asia

- India: 91-80-845-2923
- Japan: 81-120-48-2630
- China: 86-21-621-92228
- Australia: 61-2-9975-5501
- Singapore: 65-291-8528
- Taiwan: 886-2-2505-7900

GE Healthcare – Europe: 33-1-41-19-7676

GEMS on the World Wide Web: <http://www.gemedicalsystems.com/rad/mri/index.html>

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Chapter 2

GE 6-Channel Phased Array Flex Coil

Introduction

The 1.5T and 3.0T **GE 6-Channel Phased Array Flex Coil**, compatible with GE 1.5T MR450, MR450w, and 3.0T MR750 whole body MRI systems, is a receive only, multi-coil array designed to give the optimal signal-to-noise ratio (SNR) and uniformity for high resolution examinations. The coil enables you to image multi-oblique slices. The multi-element design improves the coverage and resolution.

When using the MR system for imaging using the **GE 6-Channel Phased Array Flex Coil**, refer to this manual for coil details while using existing product documentation for details of diagnostic scanning of patient.

The **GE 6-Channel Phased Array Flex Coil** topics discussed in within this chapter include:

- [Shipping contents](#)
- [Coil description](#)
- [Coil positioning](#)
- [Patient positioning](#)
- [Modes of operation](#)

Shipping contents

The following tables include the coil contents shipped with the specified catalog number.

Table 2-1: Catalog M0049SS shipping contents

Description	Part number	Quantity
1.5T 6-Channel Flex Coil Assembly	5363383	1
• 1.5T Receive Only Flex Coil	5229623	2
• 1.5T DV QD for 6-Ch Flex Coil	5363382	1
Operator manual	5371823-299	1
Service manual	5371824	1
Coil Config CD	5372128	1

Table 2-2: Catalog M0050SS shipping contents

Description	Part number	Quantity
3.0T 6-Channel Flex Coil Assembly	5363393	1
• 3.0T Receive Only Flex Coil	5229646	2
• 3.0T DV QD for 6-Ch Flex Coil	5363382-3	1
Operator manual	5371823-299	1
Service manual	5371824	1
Coil Config CD	5372128	1

Coil description

The 1.5T and 3.0T **GE 6-Channel Phased Array Flex Coil** is a receive only, multi-coil array optimized for high resolution examinations. The coil enables you to image multi-oblique slices. The coil is indicated for use, on the order of a physician in conjunction with an MR scanner, as an accessory to produce 2D and 3D images.

Intended uses

The intended areas of interest to be scanned with the **GE 6-Channel Phased Array Flex Coil** include, but are not limited to: head, neck, knee, shoulder, foot, wrist, and hip.

Coil components

The coil consists of two flexible parts enabling large opening and flexibility. Both halves are flexible in order to conform to the contours of the patient, and to accommodate the patient's anatomy. This maximizes signal integrity and clinical utility. Both coil halves are connected to the system through a Quick Disconnect box, which connects directly to the Low Profile Carriage Assembly (LPCA) of the MRI system. Coil setup flexibility allows the use of only one coil half, if desired.



CAUTION: Correctly aligning the array enables optimization of advanced parallel imaging techniques such as ASSET and ARC and provides a uniform signal across the sensitive volume, eliminating bright spots, high intensity areas and signal drop-off. Each of these halves has a cable emerging out to the point where they plug into a Quick Disconnect box, which is then connected to the LPCA of the MR system.

Figure 2-1: GE 6-Channel Phased Array Flex Coil with Quick Disconnect box



Coil positioning

To position the **GE 6-Channel Phased Array Flex Coil**, perform the following procedures for routine scanning. The coil can be used with one or both flexible paddles.



CAUTION: Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from coupling.

Routine coil positioning procedures

1. Place one paddle of the coil on the patient cradle so that it sits snugly as shown in Figure 2-2.
2. Position the patient so that there is no direct contact between the patient's skin and magnet bore or coil.
 - Use non-conductive padding that is at least 0.25 inches thick between the patient's skin and the magnet bore.
 - Use additional pads to immobilize the patient and make them comfortable.
 - Place a clean cotton sheet over the coil and comfort pad so the patient's skin does not come in contact with the coil or the comfort pad.
3. If both coil paddles will be used, place the second coil paddle around the patient's anatomy to be imaged.

Figure 2-2: Coil setup



4. Place the coil in a way such that the desired region of the patient will be centered on the coil.
 - The coil can be placed in A/P or L/R direction as preferred for the region of interest.



For instructions on coil positioning procedures with the Surgical Suite, please refer to the Surgical Suite operator manual, publication number 5310534. For coil positioning instructions with Oncology procedures, please refer to the DiscoveryTM MR750 3.0T, DiscoveryTM MR450 1.5T, OptimaTM 450w 1.5T operator manual, publication number 5375781-EN.

Patient positioning

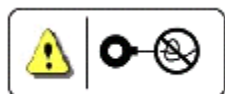
To position the patient, perform the following steps:

1. Position the patient on the posterior half of the array coil, between the two coil halves.
 - If indicated, prepare the patient with the **Respiratory Bellows** and/or **ECG leads** before proceeding.
2. With the patient lying down, gently use pads or straps as necessary to position the coil in the desired position for imaging.
 - The following figures show examples of how the coil might be set up for a subset of the applications.

Figure 2-3: 6-Channel Phased Array Flex Coil patient setup - head

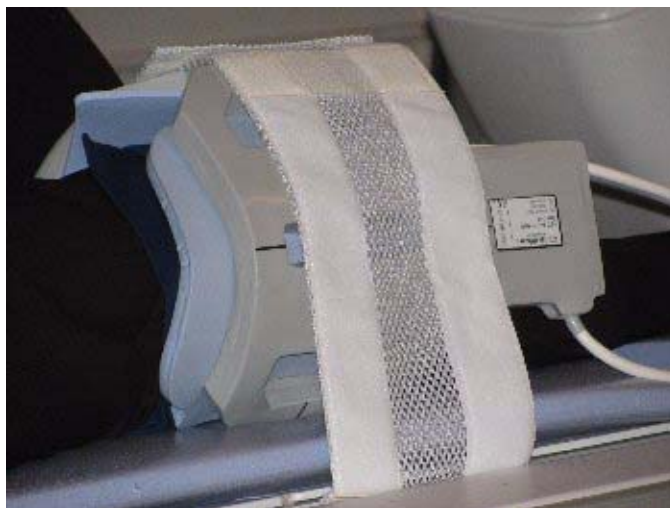


Figure 2-4: 6-Channel Phased Array Flex Coil patient setup - hip



Coil CAUTION: Make sure the coils do not touch the magnet bore. Patient burns may result from coupling.

Figure 2-5: 6-Channel Phased Array Flex Coil patient setup - knee



3. Provide earplugs for the patient after all instructions have been given.



WARNING: Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

- Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better (e.g., 30 dB, 32 dB, etc.).
4. Plug the **Coil Connectors** into the **Quick Disconnect box**.
 5. Connect the **Quick Disconnect box** into the appropriate **Phased Array Port** on the LPCA (Figure 2-6).

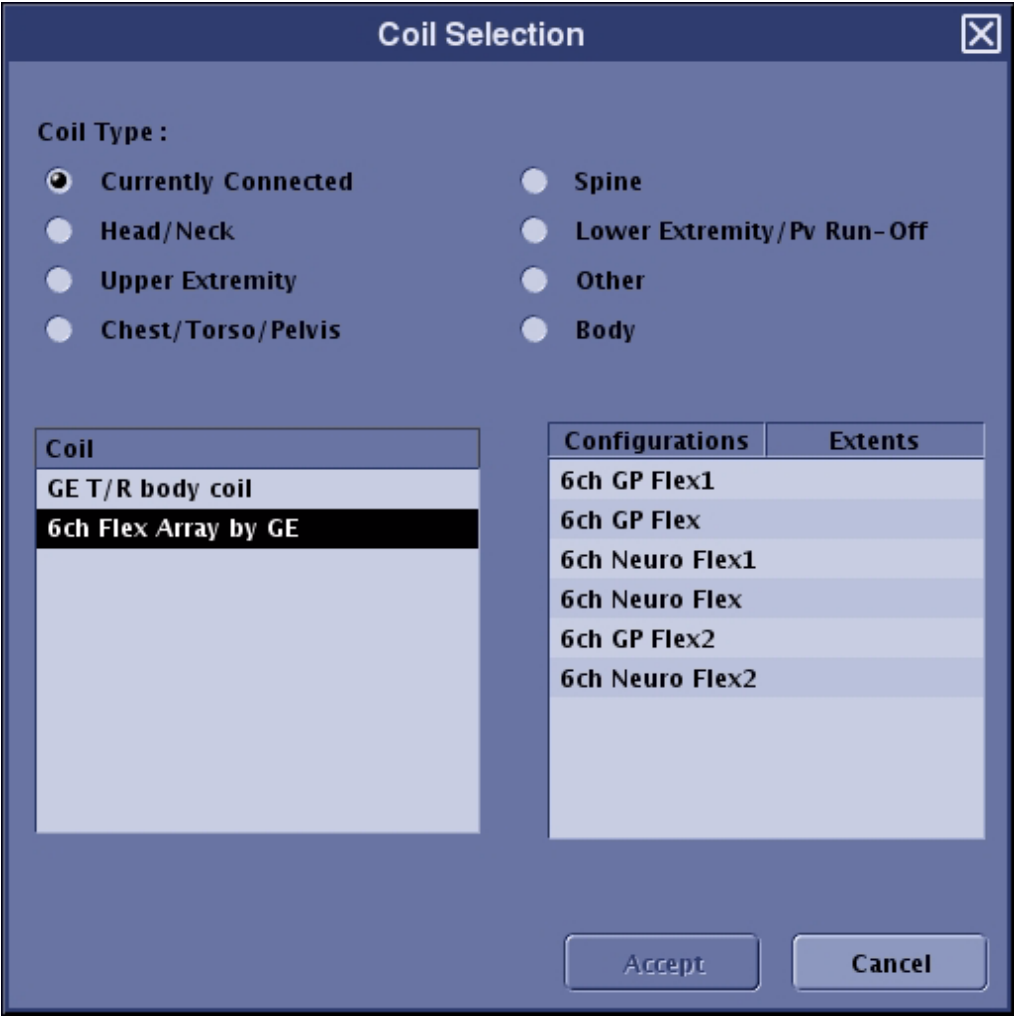
Figure 2-6: LPCA



6. Make sure that the coil cable is not looping at any point within the magnet bore.

7. Proceed with scan prescription and select the **6 Ch Flex Array by GE coil** at the coil screen (Figure 2-7).
 - With both halves of the coil connected, you have full flexibility to choose the whole coil or one half configuration.
 - The Currently Connected feature on the 1.5T MR450, MR450w, and 3.0T MR750 systems display the connected coil name and available configuration.

Figure 2-7: Coil Names screen



The image shows a 'Coil Selection' dialog box. It has a title bar with a close button. Inside, there's a 'Coil Type' section with radio buttons for 'Currently Connected' (selected), 'Head/Neck', 'Upper Extremity', 'Chest/Torso/Pelvis', 'Spine', 'Lower Extremity/Pv Run-Off', 'Other', and 'Body'. Below this are two tables. The left table is titled 'Coil' and lists 'GE T/R body coil' and '6ch Flex Array by GE' (which is highlighted). The right table has two columns: 'Configurations' and 'Extents'. It lists several configurations: '6ch GP Flex1', '6ch GP Flex', '6ch Neuro Flex1', '6ch Neuro Flex', '6ch GP Flex2', and '6ch Neuro Flex2'. At the bottom are 'Accept' and 'Cancel' buttons.

Coil
GE T/R body coil
6ch Flex Array by GE

Configurations	Extents
6ch GP Flex1	
6ch GP Flex	
6ch Neuro Flex1	
6ch Neuro Flex	
6ch GP Flex2	
6ch Neuro Flex2	



For tables with the Touch and Go functionality enabled, refer to the MR system operator manual for instructions for use.

8. Instruct the patient to close his or her eyes, and then turn the Laser Alignment Light on.
 - The Laser Alignment Light for patient positioning can cause eye injury.
9. Align the laser or localizing light so that the desired anatomy lies at the center of the coil.
 - To aid in the centering of the coil in the magnet, positioning aids have been provided on the coil in the form of dark cross hairs.
 - The localizing light should be aligned with these cross hairs for positioning accuracy within the bore.



CAUTION: Instruct the patient to close his or her eyes when landmarking and when the laser light is turned on. Exposing eyes to the laser alignment lights may result in eye injury.

10. Use the system recommended protocols for scanning with this coil for optimal results and best performance.
 - Specific choice of protocols is the responsibility of the radiologist and/or technologist.

Modes of operation

The **6-channel Flexible coil** can be operated in multiple modes. See Figure 2-8 for an understanding of the QD connector. If only one QD port is connected, then half of the coil is active and only three channels of the coil are operational to reconstruct an image.

Table 2-3: Coil modes for Neuro applications

Coil mode	Number of coil paddles	QD port number	Number of channels reconstructed
6ch Neuro Flex	2	1 and 2	6
6ch Neuro Flex1	2	1	3
6ch Neuro Flex2	2	2	3
3ch Neuro Flex1	1	1	3
3ch Neuro Flex2	1	2	3

Table 2-4: Coil modes for General Purpose applications

Coil mode	Number of coil paddles	QD port number	Number of channels reconstructed
6ch GP Flex	2	1 and 2	6
6ch GP Flex1	2	1	3
6ch GP Flex2	2	2	3
3ch GP Flex1	1	1	3
3ch GP Flex2	1	2	3

Figure 2-8: 6-Channel Flex Coil QD connectors



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Chapter 3

Maintenance

Introduction

Maintaining a controlled environment involves routine cleaning and maintenance checks of your **GE 6-Channel Phased Array Flex Coil**. Careful planning and diligent upkeep of your coil can provide a safe environment for both patients and employees.

Be aware of the cleaning guidelines and required maintenance for the following:

- [Coil and components care](#)
- [Cleaning](#)
- [Storage](#)

Coil and components care



WARNING: Detach the coil and intermediate connector from the scanner before attempting to clean. Do not reattach after cleaning until the coil and intermediate connector have dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

Cleaning

The coil, intermediate connector, and patient comfort pads must be cleaned and stored after each use using the following procedures:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- Do not spray or pour any cleaning solution directly on the coil or the intermediate connector!
- Let the coil, intermediate connector housing, and pads dry before use.
- Store the coil and the intermediate connector in an air-conditioned scan room or equipment room.

IMPORTANT: Under no circumstances should the coil or intermediate connector be placed into any type of sterilizer!

Storage

Store the coil and intermediate connector in an air-conditioned scan room or equipment room. To store the coil, a storage space of 50 cm (width) × 60 cm (depth) × 30 cm (height) is required.

Index

Numerics

- 6-ch Flex PA coil 2-1
 - coil positioning 2-4
 - description 2-2
 - maintenance 3-1
 - operation modes 2-9
 - patient positioning 2-5
 - safety 1-1

C

- caution 1-2
- cleaning 3-2
- coil
 - description 2-2
 - operation modes 2-9
 - positioning 2-4
- coil care 3-2
- coil caution 1-2
- coil modifications 1-4
- coil operating safety 1-4
- contraindications 1-5

D

- danger 1-2

E

- electrical and mechanical safety 1-6
- emergency procedures 1-5
- equipment
 - safety 1-6
- equipment safety 1-6

H

- hearing protection 2-6

I

- important 1-2
- incident reporting 1-7

M

- maintenance
 - cleaning 3-2
 - coil care 3-2
 - storage 3-2
- modes of operation 2-9

N

note 1-2

O

operation modes 2-9

P

patient

- contraindications 1-5

- positioning 2-5

- safety 1-5

protocols 2-8

Q

quality assurance 1-4

S

safety 1-1

- coil operation 1-4

- electrical 1-6

- emergency procedures 1-5

- equipment 1-6

- incident reporting 1-7

- mechanical 1-6

- patient 1-5

- patient contraindications 1-5

- symbols 1-2

storage 3-2

symbols 1-2

- caution 1-2

- coil caution 1-2

- danger 1-2

- warning 1-2

W

warning 1-2